

$^{40}\text{Ca}(\mu^-, \nu\alpha n\gamma)$ 2006Me08

2006Me08: Muons were generated from the decay of 90-MeV/c π^- beam in a 6-m, 1.2-T superconducting solenoid and were selected by a bending magnet at the M9B beamline at TRIUMF. The target was 15-g natural calcium. γ rays and x rays were detected using a 186-cm³ HPGe detector. Charged particles caused by electrons from muon decay were tagged by a plastic scintillator in front of the HPGe detector. Measured E_γ , I_γ , $E(x \text{ ray})$, $I(x \text{ ray})$. Deduced levels, muon capture yields.

Muonic Lyman series for natural Calcium

μ x ray	Energy	Intensity in percent
2p-1s	783.659 25	83.8 10
3p-1s	940.63 10	6.2 2
4p-1s	995.48 10	2.0 1
5p-1s	1020.81 10	2.0 1
6p-1s	1034.62 10	1.8 1
7p-1s	1042.71 20	1.4 1
(8 $^-\infty$)p-1s	1046-1063	2.8 4

Muonic Balmer series for natural Calcium

μ x ray	Energy	Intensity in percent
3d-2p	157.35 13	64.5 9
4d-2p	212.03 10	8.85 20
5d-2p	237.31 10	4.34 20
6d-2p	251.06 10	3.29 20
7d-2p	259.45 10	1.37 20
(8 $^-\infty$)d-2p	261-277	1.4 3

 ^{35}Cl Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\dagger$</u>
0	3/2 ⁺
1219.4	1/2 ⁺
1763.2	5/2 ⁺
2645.7	7/2 ⁺
2694.0	3/2 ⁺

[†] From Adopted Levels. Level energies are rounded-off values.

 $\gamma(^{35}\text{Cl})$

<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
1219.4	<0.3	1219.4	1/2 ⁺	0	3/2 ⁺
1763.2	0.5 3	1763.2	5/2 ⁺	0	3/2 ⁺
2645.7	<0.2	2645.7	7/2 ⁺	0	3/2 ⁺
2693.7	<0.2	2694.0	3/2 ⁺	0	3/2 ⁺

[†] Rounded-off values from Adopted Gammas.

[‡] Percent γ -ray yield per muon capture (2006Me08).

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Level Scheme

 Intensities: percent γ -ray yield per muon capture

Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$

